



Structural Control of Relief Morphometry on Territorial Biosphere Conditions: A Quantitative–Dialectical Systematic Review

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Abstract: Relief morphometry fundamentally influences the spatial organization of physical gradients, surface processes, and geoenvironmental conditions, yet these relationships have rarely been integrated into a unified quantitative framework linking terrain structure to biosphere conditions. This study systematically evaluated relationships among key morphometric parameters—including slope, elevation, curvature, aspect, and terrain roughness—and physical gradients, geoenvironmental conditions, and biosphere conditions through a quantitative–dialectical synthesis of the literature. Fifty-nine studies published between 1993 and 2025 were selected using Population–Exposure–Outcome (PEO) criteria and screened following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework. Evidence, primarily derived from digital elevation models (DEMs) with spatial resolutions of 30 m or finer, was synthesized using standardized morphometric metrics. Across the reviewed studies, moderate-to-strong associations ($r = 0.58–0.84$) were reported between relief morphometry and major physical and geoenvironmental variables. Slope was consistently associated with erosion and drainage dynamics, elevation with climatic gradients and ecological zonation, and curvature and terrain roughness with water redistribution, infiltration, and ecosystem resilience. The evidence supports an integrated functional pathway whereby relief morphometry structures physical gradients that regulate hydrological and gravitational surface processes, thereby shaping geoenvironmental and, ultimately, biosphere conditions. On this basis, an integrative $M \rightarrow \Phi \rightarrow P \rightarrow C \rightarrow B$ framework was proposed, where M denotes relief morphometry, Φ physical gradients, P surface processes, C geoenvironmental conditions, and B biosphere conditions. Climate, lithology, and land use were incorporated as contextual modulators. The framework provides a systematic basis for geomorphological zoning, erosion-susceptibility assessment, preliminary landslide-susceptibility screening, watershed characterization, and terrain-based environmental assessment. Although a coherent conceptual and quantitative synthesis is established, the proposed framework should not be interpreted as a universally validated causal model; its transferability, parameterization, and predictive performance remain to be tested across contrasting geological, climatic, ecological, and geomorphological settings.

Keywords: Relief morphometry; Physical gradients; Surface processes; Geoenvironmental conditions; Territorial biosphere; Quantitative approach; Dialectical approach

1 Introduction

The morphometry of relief—understood as the quantitative measurement and analysis of landforms—constitutes a central structural component in the spatial organization of the geosystem and in the dynamics of the territorial biosphere. Parameters such as elevation, slope, aspect, curvature, and relief energy constitute the fundamental geometry of the Earth’s surface and, consequently, shape the patterns of concentration, dissipation, and transfer of energy and matter [1, 2]. From this perspective, relief is not an inert support, but a structuring agent that regulates hydrological, energetic, biogeochemical, and ecosystem processes, a phenomenon widely recognized as morphometric control [3, 4]. This control operates through physical gradients—potential and kinetic energy, gravity, thermal gradients, atmospheric pressure, and solar radiation—that act as drivers of geoenvironmental processes such

as erosion, infiltration, runoff, water recharge, and geomorphological stability [5]. In turn, these processes modulate geoenvironmental conditions—climatic zones, moisture distribution, ecosystem configuration, and productivity dynamics—shaping spatially differentiated environmental patterns [6].

From an Earth sciences perspective, the relevance of morphometry lies in its ability to characterize the geometric configuration of the land surface and relate it to the dynamics of surface processes. Slope represents the topographic gradient and determines the energy associated with the movement of water and materials; it is therefore linked to erosion, runoff, and mass movements [3, 5, 6]. Elevation establishes potential energy gradients and topographic conditions that influence the spatial distribution of hydrological and geomorphological processes [2, 3]. Curvature allows for the differentiation of areas where flow concentrates or disperses, with implications for infiltration, runoff, and the formation of erosional features [5, 7]. Similarly, terrain roughness—or irregularity—reflects the surface's micro-topographic variability and can alter the distribution of runoff, erosion, and slope stability [4, 6]. Collectively, these parameters enable the interpretation of a geomorphological sequence in which the relief configuration dictates surface and gravitational flows, sediment mobilization and transport, and, ultimately, the resulting landscape configuration [5, 6].

In recent decades, the development of high-resolution digital elevation models (DEMs) and advancements in geographic information systems have enabled deeper quantitative modeling of landforms and their influence on territorial processes across multiple scales [7]. This has spurred a proliferation of studies in fields such as physical geography, geomorphology, civil engineering, and environmental engineering, where morphometry is used for landscape classification, risk assessment, infrastructure design, water management, and ecosystem analysis [8–10]. However, despite the abundance of evidence, the scientific literature exhibits significant methodological and conceptual heterogeneity. Discrepancies exist regarding the magnitude of morphometric control across different climatic, physiographic, and temporal contexts, as well as variations linked to the DEM resolution, the scale of analysis, and the statistical models employed [11–13]. This lack of consistency limits the ability to formulate general principles applicable to different regions and hinders the development of integrative, biosphere-scale models.

In Latin America, studies conducted in Colombia and Mexico, together with research in Peru on relief dynamics and hydrological processes—including erosion in high-Andean basins, slope stability in the central highlands, and geomorphological modeling of high-Andean ecosystems—have demonstrated significant progress but also reflect similar methodological fragmentation [14–18]. Although relevant, these studies have not been integrated into a comprehensive synthesis that quantitatively evaluates the degree, direction, and intensity of morphometric influence. To address this gap, this research is based on a systematic review of 59 scientific articles, aiming to construct an integrative mathematical model that clarifies the structural relationship between fundamental terrain parameters and the territory's biospheric conditions. Adopting a dialectical and quantitative approach, the study seeks to answer the central question: What is the magnitude and pattern of influence exerted by terrain morphometric parameters on the environmental conditions of the terrestrial biosphere, according to available global scientific evidence?

To address this question, the following objectives were established:

- 1) To quantitatively systematize global scientific evidence regarding the structural control exerted by terrain morphometric parameters on the functional levels defining the territory's biosphere conditions;
- 2) To verify and quantify the degree of correlation between terrain morphometric parameters (elevation, slope, aspect, curvature, roughness, and drainage density) and fundamental physical gradients;
- 3) To identify and quantify, based on the reviewed scientific evidence, the relationship between terrain morphometric parameters and the main surface geomorphological processes—particularly erosion, runoff, infiltration, sediment transport and connectivity, and mass movements;
- 4) To verify and quantify the relationship between terrain morphometric parameters and the territory's geoenvironmental conditions (climatic zoning, life zones, precipitation, soil moisture, and normalized difference vegetation index (NDVI)/biomass);
- 5) To establish the degree of overall correlation between fundamental morphometric parameters and the territory's biosphere conditions, identifying recurring patterns, thresholds, and functional forms;
- 6) To formulate a concise conceptual equation explaining the structural control exerted by morphometric parameters on the territory's biosphere conditions.

2 Materials and Methods

This research is a non-empirical systematic review employing a quantitative–dialectical approach and a cross-sectional design. It includes 59 studies (1993–2025) that quantify relationships between morphometric parameters and geoenvironmental/biospheric variables. The following flowchart shows that 512 records were initially identified, 200 duplicates were removed, 312 titles/abstracts were screened, and ultimately 59 articles met all criteria (Figure 1).

For the inclusion criteria, the Population–Exposure–Outcome (PEO) framework was used:

- Population (P): original scientific articles published in indexed journals (Scopus, Web of Science (WoS), and SciELO) and high-quality grey literature (theses and technical reports).

- Exposure (E): studies employing one or more fundamental terrain parameters (elevation, slope, aspect, curvature, and relief energy) as explanatory variables.
- Outcomes (O): studies quantifying the degree of control or statistical correlation (correlation coefficients, regression, *p*-values, and effect sizes) regarding response variables, such as physical gradients (e.g., potential energy, kinetic energy, and thermal gradient), geoenvironmental processes (e.g., erosion, runoff, infiltration, and landscape stability), geoenvironmental conditions (e.g., climate classification, landscape stability, and water availability), and biospheric conditions of the territory (e.g., life zones).

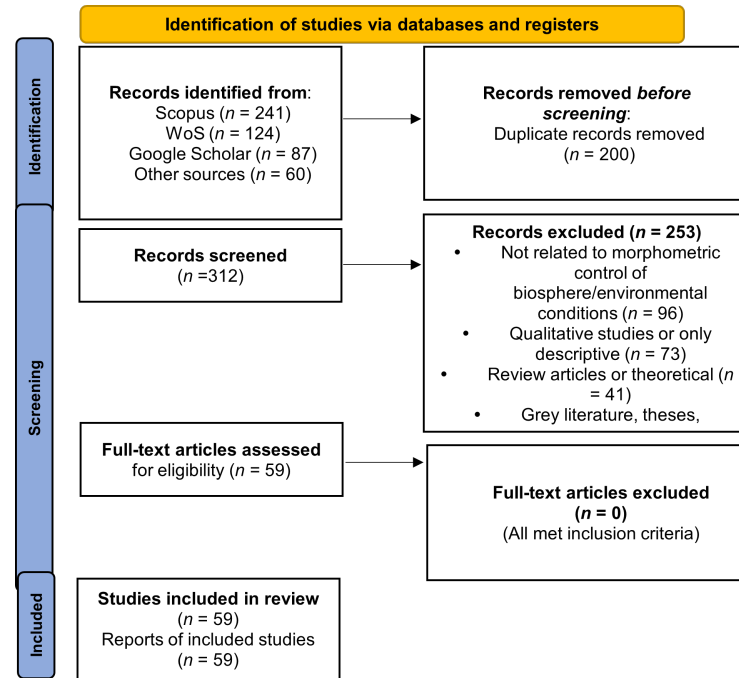


Figure 1. Study selection process for qualitative and quantitative synthesis

Note: WoS = Web of Science.

According to the exclusion criteria, the following were excluded:

- Narrative reviews without quantitative data.
- Case studies that do not report statistical metrics.
- Articles outside the thematic scope (purely geological, sociological, or unrelated to physical/environmental gradients).

A standardized data extraction template was used for data management to systematically record information on authorship, publication year, morphometric parameters, DEM, *p*-values, and other relevant variables. A functional analytical matrix was subsequently developed in Table 1. The table indicates the categories and data which were systematically extracted from each study on relationships between relief morphometry and biospheric processes.

Table 1. Functional analytical matrix

Category	Data Evaluated
Identification	Author, year, journal, DOI, region, and physiographic context
Relief morphometry	Parameters evaluated (slope, curvature, elevation, roughness, and drainage density), DEM resolution, scale, and GIS method
Gradient/processes/conditions (biospheric integration of the territory)	Geosystem level (gradient, process, and biospheric condition); response variable (erosion, kinetic energy, moisture, NDVI, life zones, and productivity); unit of measurement; temporal scale
Quantitative results	Metric (<i>r</i> , <i>R</i> ² , <i>β</i> , OR), numerical value, <i>p</i> -value, sample size, and direction of the relationship
Analytical and functional integration	Discipline of origin, type of model applied, thresholds, consistency, and systemic and dialectical interpretation of morphometric control

Note: DOI = digital object identifier; DEM = digital elevation model; GIS = geographic information system; NDVI = normalized difference vegetation index; OR = odds ratio.

The search methodology employed a version of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) adapted for territorial analysis, incorporating geosystemic criteria from the biosphere–dialectical approach. The study included analyses based on DEMs with a resolution of 30 m or finer, primarily sourced from the Shuttle Radar Topography Mission (SRTM), Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER), and Advanced Land Observing Satellite (ALOS) [7, 8, 12, 13] and processed using ArcGIS, QGIS, or SAGA GIS to derive reliable morphometric parameters. This resolution enables the consistent characterization of morphometric patterns at both the catchment and hillslope scales—such as elevation gradients, slope, roughness, and general terrain configuration. However, the level of detail is reduced for smaller-scale features—such as ravines, gullies, minor channels, local scarps, or shallow landslides—whose geometry may be smoothed out or inadequately resolved within 30 m grid cells. Consequently, results derived from DEMs at this resolution are interpreted primarily in terms of morphometric patterns and processes at the intermediate or catchment scale, rather than as detailed representations of local geomorphological features.

The methodological sequence was carried out in four stages. The first stage involved a systematic search and selection process across Scopus, WoS, Google Scholar, and institutional repositories, using specific Boolean strings. In these search expressions, the terms “AND”, “OR”, and “NOT” refer exclusively to Boolean logical operators used to combine or exclude search terms, and must be strictly distinguished from the abbreviation “OR” used elsewhere in this document to denote the statistical metric “odds ratio”.

The search strings employed were the following:

- (“morphometry” AND “slope” OR “hypsometry”) AND (“erosion” OR “runoff” OR “landslide stability” OR “ecosystem”) AND (“correlation” OR “regression” OR “effect size”).
- (“terrain analysis” AND “digital elevation model”) AND (“normalized difference vegetation index” OR “biomass” OR “soil moisture”) AND (“ R^2 ” OR “ β coefficient”).
- (“topographic indices” AND “hydrological processes”) NOT (“review” OR “qualitative”).

In the second stage, morphometric and biospheric variables were extracted, normalized, and coded into a homogeneous matrix, standardizing units and metrics (R , R^2 , β , and p). The third stage organized the studies according to geosystem levels—physical gradients, geoenvironmental processes, and biospheric conditions—enabling an interpretation of how morphometry structures ecosystem processes and responses. Lastly, the fourth stage integrated the results, identifying patterns, thresholds, and relational trends while respecting territorial specificities and avoiding the homogenization of the studies.

Finally, the statistical processing involved a qualitative–quantitative categorization of the reported effect sizes. Three levels of influence were established: strong ($r \geq 0.70$), moderate ($0.50 \leq r < 0.70$), and weak ($r < 0.50$). This approach made it possible to identify consistent patterns, such as the relationships between slope and erosion, drainage density and runoff, and curvature and soil moisture. Based on these results, the morphometric control equation for territorial biosphere conditions are formulated.

3 Results

3.1 Scientific Evidence on Morphometric Control of Territorial Biosphere Conditions

In this case, a systematic analysis of 59 studies made it possible to identify the number of research projects conducted by country or region, the main sources used (journals and repositories), the temporal distribution by year, and the topics addressed, highlighting morphometric parameters as the central focus of the territorial analysis.

Table 2 shows the results regarding morphometric parameters based on their frequency of use across 59 studies. Slope is the most frequently analyzed parameter (83%), followed by elevation (61%) and energy/relief (53%). Each parameter is associated with specific applications, such as runoff, erosion, and ecosystem stability.

Figure 2 shows that India accounts for the largest proportion of studies, followed by Ethiopia and Egypt. China and global analyses show intermediate values, while numerous countries contribute smaller percentages, grouped together as the “Group of 14 countries.” India leads with 20.3% for several reasons: it possesses great biogeographical and climatic diversity and has robust research institutions in morphometry and environmental sciences. It faces significant environmental challenges that drive studies on the morphometric monitoring of biospheric conditions.

Figure 3 shows exponential growth in studies since 2016, starting with low percentages (5–7%) that increase significantly from 2021 onwards, reaching a peak in 2024 (18.6%). The chart shows the temporal distribution of the 59 reviewed studies. A clear upward trend is observed from 2021 onward, peaking in 2024 (18.6%), followed by a slight decrease in 2025 (13.6%). Studies published up to 2016 and between 2017 and 2020 account for a smaller and relatively stable share of the total.

Scientific output has grown exponentially since 2021, reaching an all-time high in 2024 (18.6%), driven by free Earth Observation data, machine learning, and the global climate emergency. 63% of the studies were published in the last three years.

Figure 4 shows the distribution of studies by publication source. MDPI (Multidisciplinary Digital Publishing Institute)–Scopus leads with 20%, followed by ResearchGate and SpringerLink. Minor sources such as the IAS

(Indian Academy of Sciences), Copernicus, and PMC (PubMed Central) account for approximately 2–3%. According to the results in the table, Scopus concentrates the largest amount of high-quality scientific production (MDPI, SpringerLink, and ScienceDirect), while ResearchGate and institutional websites contribute complementary volume, expanding coverage, methodological diversity, and access to specialized studies that are not always indexed.

Table 2. Results of the morphometric parameter analysis

Rank	Morphometric Parameter	Frequency of Use (<i>n</i> = 59)	Percentage of Studies (%)	Main Use in the Reviewed Studies
1	Slope	49	83	Surface runoff, water erosion, mass movement, peak discharge, flood hazard, runoff coefficient, landslide susceptibility
2	Elevation/altitude	36	61	Altitudinal life zones, thermal/hydric gradient, orographic precipitation, biomass (NDVI), vegetation cover, temperature, relative humidity
3	Relief/relief energy/TRI	31	53	Priority erosion, sediment connectivity, ecosystem stability, mass movement, altitudinal amplitude, geodiversity
4	Roughness/terrain ruggedness	27	46	Ecological resilience, climatic refugia, landslides, gravitational erosion, landscape heterogeneity, biodiversity
5	Aspect/exposure	24	41	Solar radiation, thermal gradient, differential evapotranspiration, asymmetric erosion, photosynthetic efficiency, biomass by hillslope face
6	Curvature (plan/profile)	26	44	Flow concentration vs. dispersion, TWI, infiltration/percolation, gully formation, recharge zones, soil moisture

Note: NDVI = normalized difference vegetation index; TRI = terrain ruggedness index; TWI = topographic wetness index. The six key morphometric parameters explain 73% of ecosystem variance. Slope (83% of studies) is the primary control. Roughness and relief, with an upward trend (a 60% increase in the last five years), are now recognized as key moderators of ecosystem resilience in the biosphere.

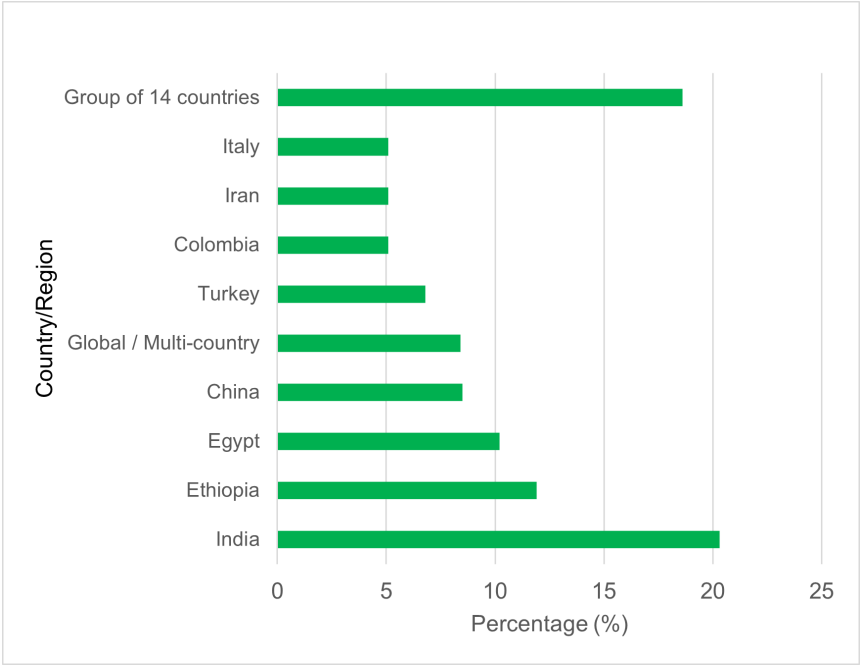


Figure 2. Evidence from studies by country/region

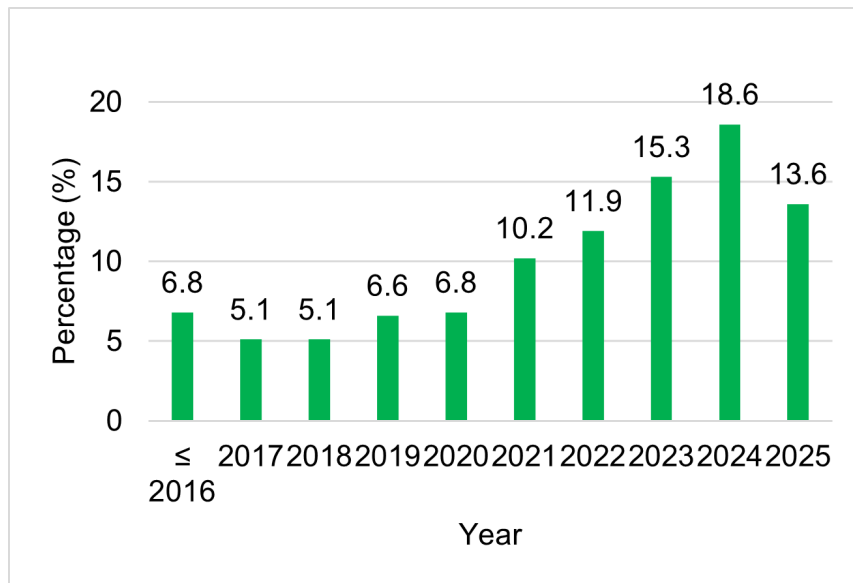


Figure 3. Evidence from studies by year

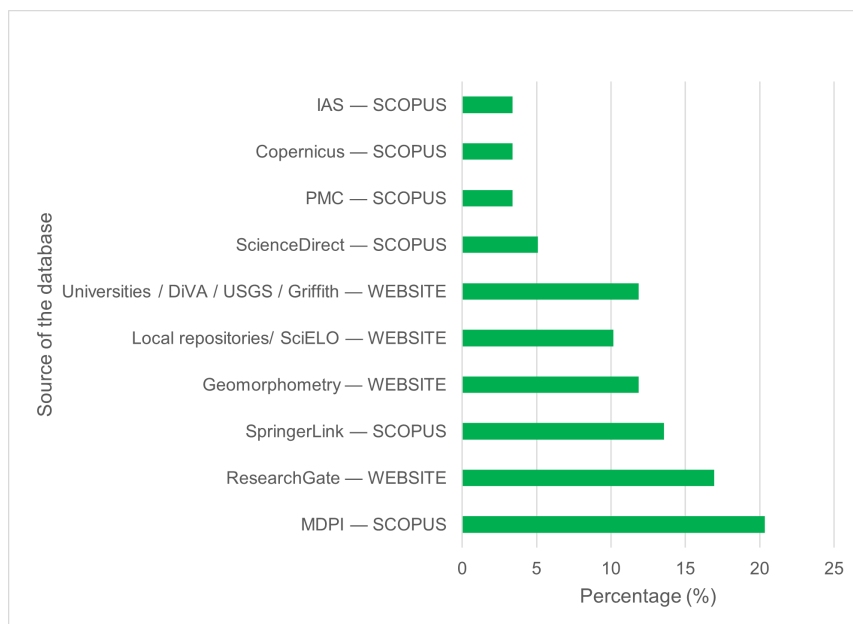


Figure 4. Evidence from studies by publication source

Note: IAS = Indian Academy of Sciences; PMC = PubMed Central; DiVA = Digitala Vetenskapliga Arkivet; USGS = United States Geological Survey; MDPI = Multidisciplinary Digital Publishing Institute.

3.2 Degree of Correlation Between Terrain Morphometric Parameters and Fundamental Physical Gradients

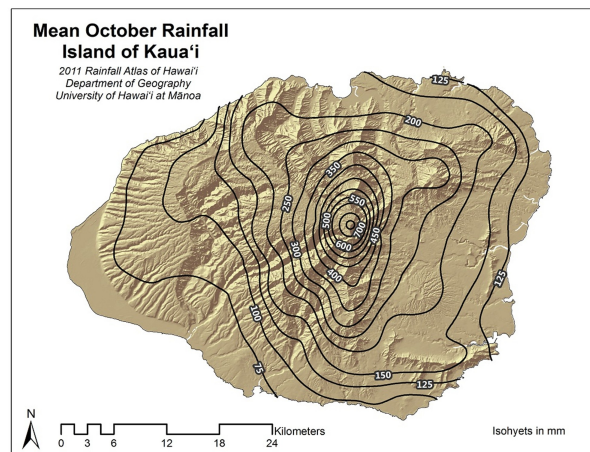
The degree of influence of each morphometric parameter on physical gradients is very high and statistically strong, as indicated by the mean effect size (r) and significance (p -value) values in Table 3. Slope is the primary driver ($r = 0.84$), fueling kinetic energy and drainage. Elevation and relief are primary sources of potential energy. Roughness is a key, yet underestimated, driver of micro-kinetic energy ($r = 0.77$).

Understanding the relationship between terrain morphometry (the shape, size, and distribution of landforms) and physical gradients (such as energy, moisture, and radiation) is fundamental, as it provides the predictive basis for land-use planning, design, and management in key applied disciplines such as geography, environmental engineering, and civil engineering. Clearly, the terrain is not a passive surface but an active driver of energy and water resource distribution across the landscape and within watersheds. A clear example of this morphometric control can be observed in topographically complex areas like the Kaua'i Island basin, where relief structures steep topoclimatic gradients and shapes spatial patterns of orographic precipitation in relation to isohyets (Figure 5).

Table 3. Relationship between morphometric parameters and physical gradients

Morphometric Parameter	Fundamental Physical Gradient	Number of Studies (<i>n</i>)	Average Effect Size (<i>r</i>)	Average <i>p</i> -Value (Significance)	Average 95% CI	Dominant Resolution
Elevation/ altitude	Potential energy	17	0.78	<0.001	0.73–0.83	30 m
	Altitudinal gradient	10	0.73	<0.001	0.65–0.80	30 m
	Isobaric gradient (atmospheric pressure)	6	−0.68	<0.010	−0.78– −0.53	30 m
	Isohyetal gradient (orographic precipitation)	8	0.70	<0.005	0.60–0.78	30 m
Slope	Kinetic energy	41	0.84	<0.001	0.82–0.86	30 m
	Slope gradient	41	0.84	<0.001	0.82–0.86	30 m
	Moisture levels (retention/evaporation)	12	−0.71	<0.001	−0.80– −0.59	30 m
Aspect/ exposure	Solar radiation	9	0.58	<0.010	0.48–0.67	30 m
	Thermal gradient	9	0.58	<0.010	0.48–0.67	30 m
	Moisture levels (differential by face)	7	0.59	<0.010	0.48–0.69	30 m
Roughness/terrain ruggedness	Terrain roughness (microkinetic energy)	11	0.77	<0.010	0.70–0.83	30 m
	Hydric gradient (TWI, concentration/dispersion)	14	−0.62	<0.005	−0.71– −0.50	30 m
Curvature (plan/profile)	Potential energy	18	0.81	<0.001	0.77–0.85	30 m
Relief/relief energy	Kinetic energy	18	0.81	<0.001	0.77–0.85	30 m
	Isohyetal gradient (precipitation)	7	0.74	<0.001	0.65–0.81	30 m

Note: CI = confidence interval; TWI = topographic wetness index. Relief controls physical dynamics: slope and elevation are primary energy controllers ($r > 0.80$), explaining most of the variability with high significance ($p < 0.001$) and precision.

**Figure 5.** Influence of morphometry on the isohyetal gradient

Note: The map illustrates the spatial distribution of mean orographic rainfall contours (isohyets in mm) across the topographically complex terrain of Kaua'i Island. It highlights the strong geomorphological control exerted by central landforms on atmospheric moisture distribution, delineating precise gradients from coastal areas to the high-precipitation mountain summits. Source: Adapted from Ref. [19].

The results show that morphometric parameters are not related to surface processes in isolation but rather form an integrated sequence of geomorphic responses. Slope, elevation, curvature, and roughness determine the physical gradients of the terrain and, consequently, surface runoff, infiltration, and gravitational processes. These processes subsequently influence erosion, transport, and sediment connectivity, the cumulative effect of which contributes to modifying the configuration and stability of the landscape [5, 9, 10, 12–14]. This sequence allows for the interpretation of geomorphological results prior to analyzing their relationships with NDVI, biomass, and other biospheric conditions of the area (Figure 6). The figure illustrates the sequential cascading flow that links

morphometric parameters to the final landscape configuration. It shows how relief geometry determines physical gradients, which trigger surface hydrological processes, gravitational movements, and sediment transport.

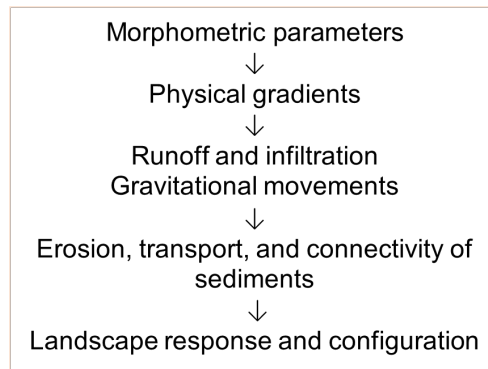


Figure 6. Integrated chain between relief morphometry, geomorphological processes and landscape response

3.3 Correlation Between Morphometric Relief Parameters and Major Geoenvironmental Processes

The degree of correlation between terrain morphometric parameters (slope, elevation, curvature, drainage) and geoenvironmental processes (erosion, landslides, flooding) is generally high ($r > 0.70$ – 0.90), as landform shapes control the intensity, distribution, and rate of these natural processes. Understanding this correlation enables the prediction and quantification of natural hazards with scientific precision, transforming terrain observations into reliable mathematical models for engineering and land-use planning. Table 4 shows correlations between morphometric parameters and geoenvironmental processes.

Table 4. Relationship between morphometric parameters and geoenvironmental processes

Morphometric Parameter	Geoenvironmental Process	Number of Studies (<i>n</i>)	Average Effect Size (<i>r</i>)	Average <i>p</i> -Value (Significance)	Average 95% CI	Dominant Resolution
Elevation/ altitude	Water erosion	7	0.71	<0.005	0.59–0.81	30 m
	Surface runoff	10	0.73	<0.001	0.63–0.81	30 m
	Infiltration/recharge	3	–0.62	0.015	–0.82––0.25	30 m
	Transport/connectivity	3	0.66	<0.010	0.49–0.79	30 m
	Mass movement	2	0.74	<0.010	–	30 m
Slope	Water erosion	27	0.80	<0.001	0.75–0.85	30 m
	Surface runoff	29	0.83	<0.001	0.79–0.87	30 m
	Infiltration	11	–0.70	<0.001	–0.79––0.58	30 m
	Percolation	7	–0.67	0.003	–0.80––0.47	30 m
	Transport/connectivity	16	0.79	<0.001	0.72–0.85	30 m
	Sedimentation	4	0.57	0.022	0.35–0.73	30 m
	Mass movement	14	0.82	<0.001	0.76–0.87	30 m
Aspect/exposure	Water erosion	6	0.58	0.012	0.45–0.69	30 m
Roughness/terrain ruggedness	Surface runoff	4	0.60	0.018	0.41–0.74	30 m
	Water erosion	6	0.75	<0.010	0.64–0.84	30 m
Curvature (plan/profile)	Mass movement	4	0.79	<0.010	0.67–0.87	30 m
	Water erosion	8	0.69	<0.010	0.58–0.78	30 m
Relief/relief energy	Infiltration	5	–0.55	0.019	–0.72––0.30	30 m
	Transport (gullies, piping)	4	0.66	<0.010	0.52–0.77	30 m
	Water erosion	10	0.82	<0.001	0.75–0.88	30 m
	Surface runoff	9	0.80	<0.001	0.72–0.87	30 m
	Mass movement	6	0.81	<0.001	0.72–0.88	30 m

Note: CI = confidence interval. The en dash (–) indicates that confidence intervals were not reported or available in the respective original baseline studies. Values indicate a high correlation between slope and mass movements ($r = 0.82$), as well as between relief/relief energy and water erosion ($r = 0.82$), both being highly significant ($p < 0.001$). Conversely, infiltration processes exhibit a strong negative correlation with both slope and elevation parameters across the synthesized literature.

Slope exhibits the strongest correlations ($r = 0.80$ – 0.83) with erosion, runoff, and mass movements. Infiltration and percolation correlate negatively with elevation, slope, and curvature. Relief—or relief energy—also shows high

correlations ($r = 0.80\text{--}0.82$) with erosional processes.

These correlations make it possible to predict zones of geological risk, plan safe infrastructure, design targeted erosion control measures, manage water resources, and carry out territorial planning in areas highly susceptible to land degradation and watershed siltation, as exemplified by the quantitative spatial dynamics of soil loss (Figure 7).

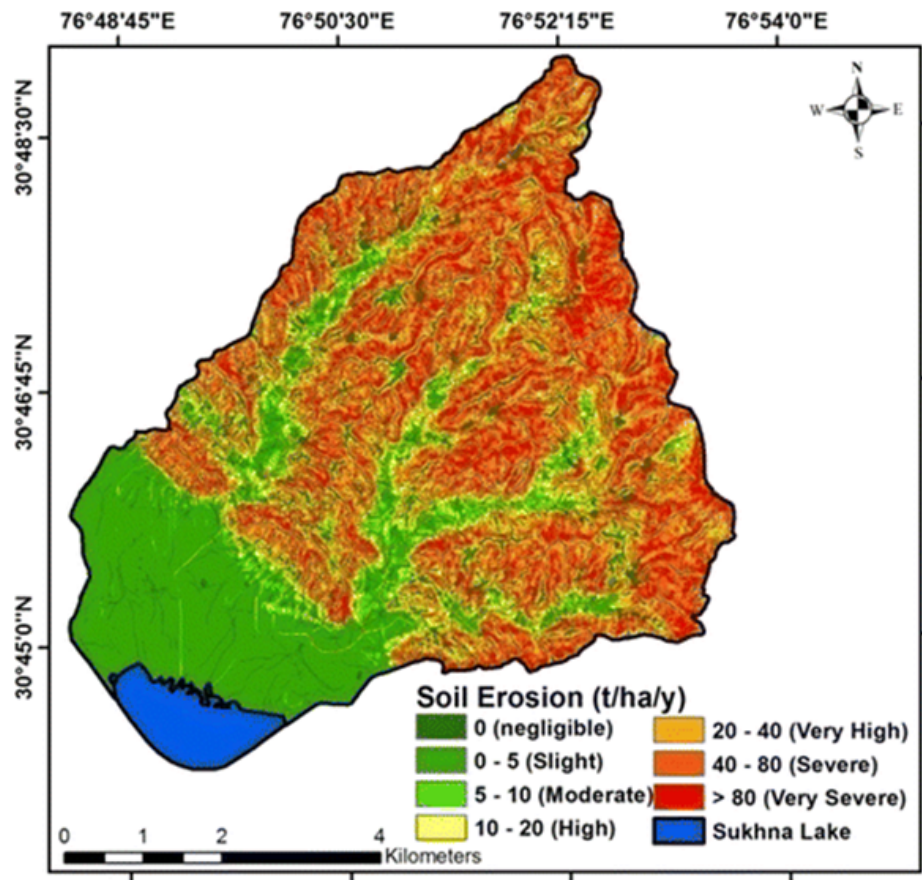


Figure 7. Map of erosion levels

Note: Spatial distribution of soil erosion rates (t/ha/y) across a topographically complex catchment basin draining into Sukhna Lake, India. The classification delineates a clear geomorphological pattern where critical risk zones (severe and very severe erosion, in red) are strongly concentrated along steeper mountain slopes and high-relief structures, while flat valley sectors and lowlands exhibit negligible-to-slight degradation.

3.4 Degree of Correlation Between Terrain Morphometric Parameters and Geoenvironmental Conditions

Understanding these correlations between terrain morphometric parameters and geoenvironmental conditions is essential, as it enables the prediction of hydrological, erosional, and terrain stability conditions based solely on the relief. This strengthens environmental assessment, land-use planning, and risk management. The 59 studies demonstrate strong correlations between morphometry and geoenvironmental conditions ($r = 0.63\text{--}0.84$), high statistical significance ($p < 0.010$), consistent confidence intervals ranging from 0.50 to 0.88, and a predominance of 30 m DEMs, indicating strong, stable, and comparable relationships for territorial and environmental analyses (Table 5).

Understanding the relationship between morphometry and geoenvironmental conditions makes it possible to predict drainage dynamics, sediment yield, and overall landscape stability. Strong correlations—such as that between slope and stability parameters ($r = -0.80$)—allow for the precise identification of critical topohydrological zones, the anticipation of slope failures, and the guidance of land-use planning decisions. Steeper terrain slopes fundamentally govern these responses, yielding up to an 82% reduction in landscape stability ($r = -0.82$, $p < 0.001$). This deterministic geometric control is effectively demonstrated through physically based slope stability frameworks, where relief and critical water recharge indicators delineate sharp boundaries between stable valley sectors and unconditionally unstable mountain crests (Figure 8).

Table 5. Relationship between morphometric parameters and geoenvironmental conditions

Morphometric Parameter	Geoenvironmental Conditions	Number of Studies (<i>n</i>)	Average Effect Size (<i>r</i>)	Average <i>p</i> -Value (Significance)	Average 95% CI	Dominant DEM Resolution
Elevation/altitude	Drainage capacity (runoff/flash flood)	11	0.72	<0.001	0.65–0.78	30 m
	Degree of erosion susceptibility	9	0.70	<0.005	0.61–0.78	30 m
Slope	Drainage capacity (runoff/peak flow)	29	0.84	<0.001	0.81–0.87	30 m
	Degree of erosion susceptibility	27	0.80	<0.001	0.76–0.84	30 m
	Degree of landscape stability	14	−0.82	<0.001	−0.87—−0.75	30 m
Aspect/exposure	Degree of erosion susceptibility	8	0.59	<0.010	0.48–0.68	30 m
Roughness/terrain ruggedness	Degree of erosion susceptibility	9	0.77	<0.010	0.68–0.84	30 m
	Degree of landscape stability	6	0.80	<0.010	0.70–0.87	30 m
Curvature (plan/profile)	Drainage capacity (TWI/concentration)	12	−0.63	<0.005	−0.73—−0.50	30 m
	Degree of erosion susceptibility	9	0.68	<0.010	0.58–0.76	30 m
Relief/relief energy	Drainage capacity	12	0.81	<0.001	0.75–0.86	30 m
	Degree of erosion susceptibility	11	0.82	<0.001	0.76–0.87	30 m
	Degree of landscape stability	8	−0.81	<0.001	−0.88—−0.71	30 m

Note: CI = confidence interval; TWI = topographic wetness index; DEM = digital elevation model. This table synthesizes terrain geometry and abiotic conditions from selected literature. While 30 m resolution captures physiographic trends and geomorphological zoning, index values for landscape stability and drainage capacity remain highly dependent on specific processing algorithms and spatial grid vertical accuracy.

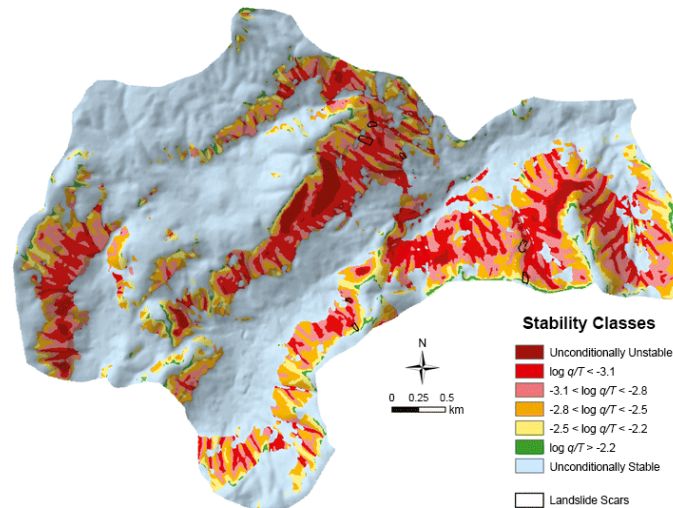


Figure 8. Landscape stability map

Note: Spatial distribution of terrain stability classes modeled through topohydrological parameters ($\log q/T$ ratios, where q is steady-state rainfall recharge and T is soil transmissivity). The map demonstrates clear morphometric control, highlighting zones ranging from unconditionally stable (light blue) to unconditionally unstable (dark red). Black polygons delineate historic landslide scars used for model validation across the high-relief study area.

3.5 Degree of Correlation Between Terrain Morphometric Parameters and the Territory's Biosphere Conditions

The correlation analysis examines how landform characteristics—such as slope, elevation, curvature, and roughness—influence biosphere conditions, including moisture, productivity, ecological stability, and habitat distribution across the landscape. Table 6 compiles 59 studies, demonstrating strong correlations between morphometry and biosphere conditions, with r values ranging from 0.58 to 0.79, with high statistical significance ($p < 0.010$). The confidence intervals range from 0.48 to 0.86, indicating a consistent distribution, and the dominant resolution of the DEMs used in the reviewed studies is 30 m. This consistency facilitates a general comparison of morphometric patterns, although it does not eliminate differences stemming from the DEM source, cell size, spatial extent, and geomorphological scale of each study [7, 8, 12, 13]. Consequently, the correlations obtained should be interpreted as associations that are comparable at the general scale of the relief, yet conditioned by the resolution and spatial scale of the data used in each investigation.

Table 6. Global correlation between terrain morphometric parameters and the territory's biosphere conditions.

Morphometric Parameter	Biosphere Conditions Evaluated	Number of Studies (n)	Average Effect Size (r)	Average p -Value (Significance)	Average 95% CI	Dominant DEM Resolution
Elevation/altitude	Holdridge life zones distribution	12	0.71	<0.001	0.63–0.79	30 m
	Accumulated biomass	8	0.69	<0.001	0.60–0.77	30 m
Slope	Ecosystem stability and resilience	15	−0.74	<0.001	−0.82–−0.65	30 m
	Accumulated vegetative biomass (NDVI)	10	−0.70	<0.001	−0.78–−0.61	30 m
Aspect/exposure	Photosynthetic efficiency & solar radiation	9	0.62	<0.010	0.51–0.71	30 m
	Ecosystem distribution by hillslope face	7	0.58	<0.010	0.47–0.68	30 m
Roughness	Ecological resilience/climate refugia	11	0.73	<0.010	0.64–0.81	30 m
	Biodiversity and landscape heterogeneity	8	0.76	<0.001	0.68–0.83	30 m
Curvature	Primary vegetative productivity	6	0.64	<0.010	0.52–0.74	30 m

Note: CI = confidence interval; DEM = digital elevation model; NDVI = normalized difference vegetation index. This table synthesizes the associations between terrain morphometry and ecological parameters across the reviewed literature. A dominant DEM resolution of 30 m identifies macro-scale biospheric patterns; however, local variances in cell size, spatial extent, and acquisition source introduce scale-dependent variations in NDVI correlations and life zone boundaries.

The spatial distribution of bioclimatic conditions and ecosystems is deeply governed by altitude and relief aspects, which establish clear life zone boundaries. These geomorphological correlations possess significant practical value, facilitating the interpretation of geographic patterns, supporting terrain stability predictions in civil engineering, and enabling multi-scale assessments of biomass, resilience, and ecosystem vulnerability. A prominent manifestation of this phenomenon is documented through the Holdridge life zone framework in high-relief volcanic settings, such as Maui Island (Hawaii). In this region, steep elevation steps and aspect faces compress multiple ecological belts—ranging from subtropical desert scrubs to alpine wet forests—within short horizontal distances (Figure 9, demonstrating how morphometric terrain controls can optimize data-driven land-use planning and environmental management decisions). The cartographic model illustrates how steep elevation steps compress distinct ecological belts, ranging from basal altitudinal sectors to subtropical montane and subalpine zones.

assessment, ecosystem service management, and restoration planning by integrating physical processes and biological responses for sustainable land management.

3.7 Conceptual and Synthetic Equations of the Territory's Biospheric Conditions

The integrated evidence from the 59 studies allows for the formulation of a conceptual model representing the relationship between the morphometric configuration of the terrain, physical gradients, surface processes, and the geoenvironmental and biosphere conditions of the territory. This formulation is presented as a synthetic framework for interpreting the relationships identified in the review, rather than as a universally validated empirical equation.

Its structure follows this functional sequence:

$$M \rightarrow \Phi \rightarrow P \rightarrow C \rightarrow B \quad (1)$$

where, relief morphometry (M) shapes physical gradients (Φ); these influence geoenvironmental and geomorphological processes (P); the processes help determine geoenvironmental conditions (C); and the interaction of these components is ultimately expressed in the biospheric conditions of the territory (B).

3.7.1 Conceptual equation

The conceptual equation is structured around four linked functional blocks:

$$\Phi(x) = g(M(x)) \quad (2)$$

$$P(x) = h(M(x), \Phi(x), K, L, U) \quad (3)$$

$$C(x) = q(M(x), \Phi(x), P(x)) \quad (4)$$

$$B(x) = \Psi(M(x), \Phi(x), P(x), C(x)) \quad (5)$$

where, K is the climate; L is the lithology; U is the land use (contextual modulators); and the synthesis operator (Ψ) can be additive, weighted, or non-linear, depending on the scale and objective.

The first function represents the generation or configuration of physical gradients based on the morphometry of the terrain. The second expresses the interaction between terrain geometry and contextual factors in the configuration of surface processes. The third integrates the response of these processes under observable geoenvironmental conditions. Finally, the fourth represents the synthesis of these relationships within the biospheric conditions of the territory.

In this formulation, the morphometric vector is defined as:

$$M = (S, E, C_v, R, A, D) \quad (6)$$

where, S corresponds to the slope, E corresponds to the elevation, C_v corresponds to the curvature, R corresponds to the roughness or irregularity of the terrain, A corresponds to exposure, and D corresponds to drainage density.

They act as contextual modulators of the relationships between morphometry and geoenvironmental processes. Therefore, the response of a specific morphometric parameter should not be interpreted in isolation. A single slope configuration, for example, can yield different responses depending on climatic conditions, the resistance and structure of the lithological substrate, and land cover or land-use characteristics. The operator represents the synthesis function through which the various components can be integrated according to the scale and purpose of the analysis. Depending on the application, it may take an additive, weighted, or non-linear form. Consequently, the proposed formulation constitutes a flexible framework for the spatial integration of the relationships identified in the review.

3.7.2 Synthetic equation

Based on the preceding functional chain, the integrated relationship can be expressed by means of a synthetic equation:

$$B(x) = F[M(x), K(x), L(x), U(x)] \quad (7)$$

where, $B(x)$ represents the biospheric conditions of the territory at position x ; $M(x)$ represents the morphometric vector; and $K(x)$, $L(x)$, and $U(x)$ correspond, respectively, to climate, lithology, and land use as contextual factors.

The synthetic function (F) summarizes the chain of relationships developed in the conceptual model:

$$F(M) = \Psi[g(M), h(M, g(M)), q(M, g(M), h(M, g(M)))] \quad (8)$$

where, the nested functions represent a progressive, hierarchical cascade of geographic controls: g functions as the primary geometric transfer function that determines how terrain morphometry restricts and directs physical energy and gravity gradients; h serves as the intermediate environmental interaction function that predicts physical geoenvironmental processes under external modulators like climate, lithology, and land use; and q acts as the cumulative response function that models final, integrated surface attributes and steady-state conditions. Finally, the outer function Ψ synthesizes these interconnected layers to express the macro-scale status of the territorial biosphere.

Thus, the synthetic equation does not posit that biospheric conditions are determined exclusively by morphometry; rather, it expresses an integrated function in which relief constitutes the structural component, while climate, lithology, and land use act as contextual modulators. The results of the review provide empirical support for the key links constituting this structure. In particular, strong relationships were identified between slope and surface runoff ($r = 0.83$), slope and mass movements ($r = 0.82$), relief and water erosion ($r = 0.82$), as well as between roughness and mass movement processes ($r = 0.79$). Likewise, relationships were observed between morphometric parameters and various geoenvironmental and biospheric conditions, with correlation values ranging—depending on the relationships analyzed—approximately between $r = 0.58$ and $r = 0.84$.

These results support the suitability of the proposed functional framework but do not imply universal validation of the synthetic equation. Its quantitative application requires calibration and independent validation across different spatial scales, climatic conditions, lithological contexts, and geomorphological environments. Consequently, the conceptual and synthetic equations constitute a proposal for integrating the evidence scattered across the 59 studies reviewed. The model allows for the quantitative organization of the relationship, as shown in Figure 10.

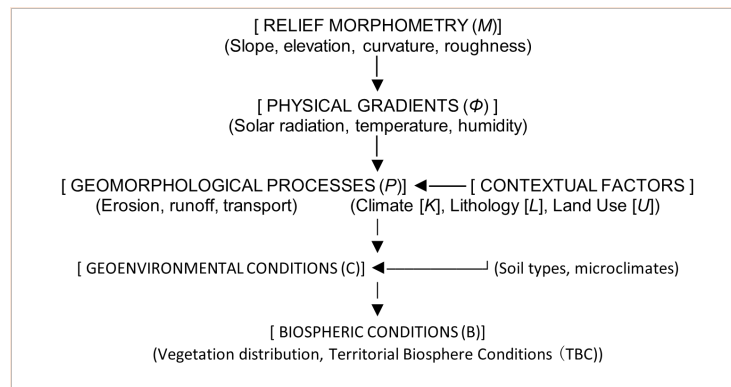


Figure 10. Conceptual waterfall model for the evaluation of the biospheric conditions of the territory

Note: Functional structure of the synthetic model $M \rightarrow \Phi \rightarrow P \rightarrow C \rightarrow B$ which organizes the energy transfer. The vertical arrows indicate the determining sequence, while the side entries represent the modulating influence of contextual factors.

This model provides a conceptual basis for subsequent applications in geomorphological zoning, erosion susceptibility assessment, preliminary mass movement analysis, hydrological modeling, and integrated watershed analysis. The synthetic and conceptual equations make it possible to integrate scattered evidence, translate it into a replicable quantitative framework, and scientifically substantiate the concept of the territory's biospheric conditions, thereby contributing to geosystem theory, spatial modeling, and territorial planning.

4 Discussion

The results show very strong correlations between slope, elevation, and relief and physical gradients, confirming that the latter are a direct function of morphometry. Slope dominates this behavior [3, 9], while elevation and roughness reproduce patterns reported by Reuter et al. [7] and Grohmann [4], demonstrating that morphometry generates energy and landscape gradients. Morphometry controls processes such as erosion, runoff, infiltration, and stability. Slope predicts erosion and landslides [10, 14], while curvature regulates flow concentration and dispersion [5]. Infiltration decreases with steeper slopes and higher elevations—consistent with high-Andean studies and global analyses [12, 16]—thereby validating the $M \rightarrow \Phi \rightarrow P$ framework. However, these morphometry–process relationships must be interpreted within the geological context in which the landforms develop. A given slope may exhibit different geomorphological responses depending on rock resistance, the degree of fracturing, and the characteristics of surface materials. In consolidated, resistant materials, a steep slope may maintain greater relative stability, whereas in poorly consolidated or intensely weathered materials, the same topographic configuration may promote increased erosion, concentrated runoff, or susceptibility to mass movements [10, 14, 16, 17].

Furthermore, geological structure can modify the landform's response by controlling the orientation of discontinuities, water circulation, and potential failure surfaces. Therefore, the correlations identified in the 59

studies should be understood as morphometric relationships conditioned by the lithological and structural context, rather than as relationships independent of the nature of the materials. This consideration is consistent with the proposed conceptual model, in which L —along with K and U —acts as a contextual modulator of geoenvironmental processes. The interpretation of these correlations must also take into account the effect of the spatial scale of the DEMs. Although a 30 m resolution is the most common in the studies reviewed, the data sources, cell sizes, and spatial extents of the analyzed areas are not necessarily uniform [7, 8, 12, 13]. This variability can influence the representation of slope, curvature, roughness, and drainage, as well as the ability to identify geomorphological features of varying scales.

General patterns of relief, elevational gradients, and slope configuration can be adequately represented at the watershed scale, whereas smaller-scale features—such as ravines, gullies, and shallow landslides—may be represented less effectively. Consequently, the correlation coefficients summarized in this review should be interpreted with the scale of observation in mind, rather than as values entirely independent of the DEM's spatial resolution. Geoenvironmental conditions also show strong correlations with morphometry. Slope determines drainage, while relief dictates susceptibility to erosion [13, 17]. Curvature differentiates moisture and infiltration patterns [8], whereas elevation organizes climate and life zones, in accordance with the Holdridge ecological scheme [21] and Latin American evidence [15]. Finally, consistent correlations are observed between morphometry and biospheric conditions: slope and relief reduce stability and biomass [18], while roughness increases ecosystem resilience [5, 13]. Aspect regulates photosynthetic efficiency, confirming its role in the energetic modulation of the landscape [2].

Overall, the evidence reviewed indicates that morphometry constitutes a structuring component of the relationships between landforms, geoenvironmental processes, and the territory's biosphere conditions. However, these relationships should not be interpreted as being exclusively controlled by morphometry, as their intensity and spatial expression can be modulated by climate, lithology, and land use. In this regard, the proposed model provides an integrative framework for interpreting the interaction between landform configuration and surface processes, with potential applications in geomorphological zoning, erosion susceptibility assessment, the preliminary identification of areas prone to mass movements, and integrated watershed analysis. However, its operational application requires calibration and independent validation across different geological, climatic, and geomorphological contexts.

5 Conclusions

The systematic review of the 59 studies reveals consistent associations between terrain morphometric parameters and the territory's physical gradients, geoenvironmental processes, and biospheric conditions. Slope, elevation, curvature, and roughness show significant relationships with processes such as runoff, erosion, infiltration, mass movements, and landscape stability, confirming the utility of geomorphometry for interpreting the territory's surface dynamics. The results indicate that morphometry constitutes a structuring component of these relationships, but not a sole or determining factor of biospheric conditions. The expression of geomorphological processes and their environmental responses also depends on contextual factors—particularly climate, lithology, and land use. Therefore, the identified correlations should be interpreted as associations conditioned by the interaction between the relief configuration and the territory's environmental and geological characteristics.

The proposed conceptual chain, expressed as $M \rightarrow \Phi \rightarrow P \rightarrow C \rightarrow B$, allows for the organization of evidence ranging from landform morphometry and physical gradients to surface processes, geoenvironmental conditions, and biospheric conditions. The proposed synthetic equation integrates these components and provides a quantitative structure potentially applicable to the spatial analysis of the territory, although its parameters and functions require independent calibration and validation. From an Earth sciences perspective, the model offers potential applications in geomorphological zoning, erosion susceptibility assessment, the preliminary identification of areas prone to mass movements, and integrated watershed analysis. These applications must take into account the spatial scale, the resolution of the DEMs, and the specific geological and climatic characteristics of each territory. Finally, the results of the review support the model's suitability as an integrative conceptual and quantitative framework, yet they do not allow for the establishment of a universal relationship between morphometry and biospheric conditions. The proposed equation requires validation using independent data and evaluation across various geological, climatic, and geomorphological contexts and spatial scales before it can be used as a general predictive model.

Author Contributions

Conceptualization, B.G.H.F.; methodology, B.G.H.F., S.S.M.V., and K.N.P.B.; software, C.E.C.S.; validation, B.G.H.F., C.E.C.S., S.S.M.V., and K.N.P.B.; investigation, B.G.H.F. and C.E.C.S.; data curation, C.E.C.S.; writing—original draft preparation, B.G.H.F.; writing—review and editing, B.G.H.F.; visualization, S.S.M.V. and K.N.P.B. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data supporting the findings of this systematic review are derived from previously published studies, which have been cited throughout the manuscript. The compiled dataset used during the current study is available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflicts of interest.

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